

ASSESSMENT -1:

Commands:

```
docker run -d -p 8080:8080 --name airflow apache/airflow standalone
```

```
docker ps - here you will see the port number
```

```
docker logs airflow
```

```
docker exec -it airflow bash
```

```
cd dags
```

```
cat>employee_salary_dag.py
```

Code:

```
from airflow import DAG
```

```
from airflow.operators.python import PythonOperator
```

```
from datetime import datetime
```

```
EMPLOYEE_FILE="/tmp/employees.txt"
```

```
SALARY_REPORT="/tmp/salary_report.txt"
```

```
def create_employee_file():
```

```
    with open(EMPLOYEE_FILE,"w") as file:
```

```
file.write("Rahul,45000\nSneha,52000\nAmit,61000\nPriya,47000\nKiran,39000")
```

```
def read_employee_data():
```

```
    with open(EMPLOYEE_FILE,"r") as file:
```

```
        data=file.readlines()
```

```
    for line in data:
```

```
        print(line.strip())
```

```

def calculate_salary_expense(ti):
    with open(EMPLOYEE_FILE,"r") as file:
        data=file.readlines()

    total_salary=0

    for line in data:
        name,salary=line.strip().split(",")
        total_salary+=int(salary)

    print("Total Salary Expense =",total_salary)

    ti.xcom_push(key="total_salary",value=total_salary)
    ti.xcom_push(key="total_employees",value=len(data))

def find_highest_salary(ti):
    with open(EMPLOYEE_FILE,"r") as file:
        data=file.readlines()

    highest_salary=0
    highest_employee=""

    for line in data:
        name,salary=line.strip().split(",")
        salary=int(salary)

        if salary>highest_salary:
            highest_salary=salary

```

```
highest_employee=name
```

```
print("Highest Salary =",highest_salary)
```

```
print("Employee =",highest_employee)
```

```
ti.xcom_push(key="highest_salary",value=highest_salary)
```

```
ti.xcom_push(key="highest_employee",value=highest_employee)
```

```
def generate_salary_report(ti):
```

```
total_salary=ti.xcom_pull(task_ids="calculate_salary_expense",key="total_salary")
```

```
total_employees=ti.xcom_pull(task_ids="calculate_salary_expense",key="total_employees")
```

```
with open(SALARY_REPORT,"w") as file:
```

```
    file.write(f"Employee Salary Report\nTotal Employees =  
{total_employees}\nTotal Salary Expense = {total_salary}\nStatus =  
Processed Successfully")
```

```
default_args={  
    "owner":"airflow"  
}
```

```
with DAG(  
    dag_id="employee_salary_processing",  
    default_args=default_args,  
    start_date=datetime(2024,1,1),
```

```
schedule_interval=None,  
catchup=False  
) as dag:
```

```
create_employee_file_task=PythonOperator(  
    task_id="create_employee_file",  
    python_callable=create_employee_file  
)
```

```
read_employee_data_task=PythonOperator(  
    task_id="read_employee_data",  
    python_callable=read_employee_data  
)
```

```
calculate_salary_expense_task=PythonOperator(  
    task_id="calculate_salary_expense",  
    python_callable=calculate_salary_expense  
)
```

```
find_highest_salary_task=PythonOperator(  
    task_id="find_highest_salary",  
    python_callable=find_highest_salary  
)
```

```
generate_salary_report_task=PythonOperator(  
    task_id="generate_salary_report",  
    python_callable=generate_salary_report  
)
```

```
create_employee_file_task>>read_employee_data_task>>calculate_salary_expense_task>>find_highest_salary_task>>generate_salary_report_task
```

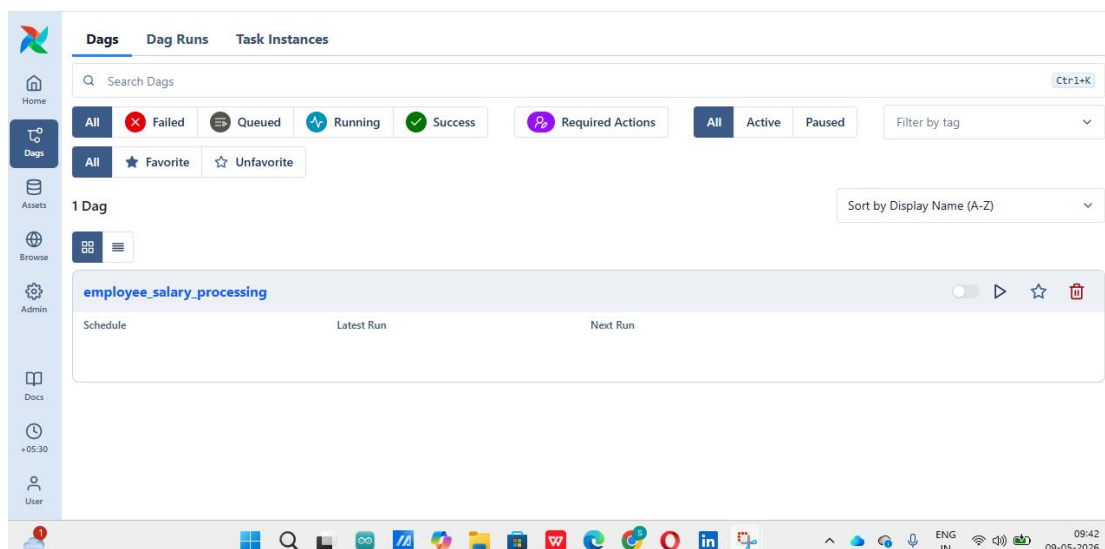
ls:

```
airflow@a6550b171b3b:/opt/airflow/dags$ ls
__pycache__  employee_salary_dag.py
```

Exit

docker restart airflow

Output screenshots:



Home

Dags

Assets

Browse

Admin

Docs

+05:30

User

Dags

Dag Runs

Task Instances

Search Dags

Ctrl+K

All

Failed

Queued

Running

Success

Required Actions

All

Active

Paused

Filter by tag

All

Favorite

Unfavorite

1 Dag

Sort by Display Name (A-Z)

employee_salary_processing

Toggle

Play

Star

Trash

Schedule	Latest Run	Next Run
	2026-05-09 09:43:52	

Home

Dags

Assets

Browse

Admin

Docs

+05:30

User

Dag

employee_salary_processing

00:00:08.656

00:00:04.328

create_employee_file

read_employee_data

calculate_salary_expense

find_highest_salary

generate_salary_report

Search Dags

Ctrl+K

Trigger

employee_salary_processing

Toggle

Star

Copy

Trash

Schedule	Latest Run	Next Run	Max Active Runs	Owner	Tags	Latest Dag Version
	2026-05-09 09:43:52		1 of 16	airflow		v1

Overview

Runs

Tasks

Calendar

Audit Log

Code

Details

Last 24 Hours

2026-05-08 09:44:27 - 2026-05-09 09:44:27

0 Failed Tasks

0 Failed Runs

Last Dag Run

Duration

00:00:08

00:00:04

0

09:43:52

Run After

All Log Levels

All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=employee_salary_processing/run_id>manual__2026-05-09T04:13:52.942119+00:00"]

0 [2026-05-09 09:43:57] INFO - DAG bundles loaded: dags-folder

1 [2026-05-09 09:43:57] INFO - Filling up the DagBag from /opt/airflow/dags/employee_salary_dag.py

2 [2026-05-09 09:43:57] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.provide' instead.

3 [2026-05-09 09:43:57] INFO - Rahu1,45000

4 [2026-05-09 09:43:57] INFO - Sneha,52000

5 [2026-05-09 09:43:57] INFO - Amit,61000

6 [2026-05-09 09:43:57] INFO - Done, Returned value was: None

7 [2026-05-09 09:43:57] INFO - Priya,47000

8 [2026-05-09 09:43:57] INFO - Kiran,39000

Dag **employee_salary_processing** **Dag Run** **manual_2026-05-09T04:13:52.942119+00:00** **Task** **calculate_salary_expense** **Search Dags** **Ctrl+K** **Trigger**

calculate_salary_expense **Success**

Operator	Start Date	End Date	Duration	Dag Version
PythonOperator	2026-05-09 09:43:58	2026-05-09 09:43:58	00:00:00.203	v1

Logs **Rendered Templates** **XCom** **Asset Events** **Audit Log** **Code** **Details**

All Log Levels All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=employee_salary_processing/run_id=manual_2026-05-09T04:13:52.942119+00:00/"]

```

0 [2026-05-09 09:43:58] INFO - DAG bundles loaded: dags-folder
1 [2026-05-09 09:43:58] INFO - Filling up the DagBag from /opt/airflow/dags/employee_salary_dag.py
2 [2026-05-09 09:43:58] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers
3 [2026-05-09 09:43:58] INFO - Total Salary Expense = 244000
4 [2026-05-09 09:43:58] INFO - Done. Returned value was: None

```

...killercoda.com/dags/.../manual_2026-05-09T04:13:52.942119+00:00

Dag **employee_salary_processing** **Dag Run** **manual_2026-05-09T04:13:52.942119+00:00** **Task** **find_highest_salary** **Search Dags** **Ctrl+K** **Trigger**

find_highest_salary **Success**

Operator	Start Date	End Date	Duration	Dag Version
PythonOperator	2026-05-09 09:43:59	2026-05-09 09:43:59	00:00:00.166	v1

Logs **Rendered Templates** **XCom** **Asset Events** **Audit Log** **Code** **Details**

All Log Levels All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=employee_salary_processing/run_id=manual_2026-05-09T04:13:52.942119+00:00/"]

```

0 [2026-05-09 09:43:59] INFO - DAG bundles loaded: dags-folder
1 [2026-05-09 09:43:59] INFO - Filling up the DagBag from /opt/airflow/dags/employee_salary_dag.py
2 [2026-05-09 09:43:59] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers
3 [2026-05-09 09:43:59] INFO - Highest Salary = 61000
4 [2026-05-09 09:43:59] INFO - Employee = Amit
5 [2026-05-09 09:43:59] INFO - Done. Returned value was: None

```

...killercoda.com/dags/.../manual_2026-05-09T04:13:52.942119+00:00

```

airflow@dddcff0de5d98:/opt/airflow$ cat /tmp/employees.txt
Rahul,45000
Sneha,52000
Amit,61000
Priya,47000

```

```

Employee Salary Report
Total Employees = 5
Total Salary Expense = 244000
Status = Processed Successfullyairflow@dddcff0de5d98:/opt/airflow$

```